

Predicting per-contact, per-gene horizontal transfer of antimicrobial resistance genes in an agent-based simulation

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Abstract

Horizontal gene transfer moves antimicrobial resistance genes between bacteria, but individual transfer events are hard to observe directly in natural communities, which makes supervised learning on them difficult to set up. We use an agent-based simulation, in which every transfer is recorded as it happens, to pose a task with exact labels: given a snapshot of a bacterial population, predict for each directed cell-to-cell contact and each resistance gene whether that gene is transferred in the next time window. We train a graph attention network on this task and evaluate it against per-gene random forest and logistic regression baselines.

The network reaches a test AUROC of 0.9777 ± 0.0023 on the four genes our per-gene random forest can fit in every seed, against 0.9548 ± 0.0124 for that baseline, and is ahead on all five model seeds; its macro average over the eight genes in our evaluation policy is 0.9805 ± 0.0012 . Because it is trained jointly on all gene outputs, it also gives stable predictions for genes the baselines cannot reliably fit at all: two that fall below their five-positive training threshold in every seed, and a third that falls below it in three seeds of five. We attribute this coverage advantage to parameter sharing across a multi-label output rather than to the graph: removing message passing costs only 0.0031 ± 0.0014 , and the genes a cell already carries dominate every other feature group by an order of magnitude.

We are explicit about what these figures do not establish. The simulator transfers genes only between cells of the same species, so the interspecies transfers that dominate the literature on these genes are absent from the task, the labels and the evaluation alike; in our dataset only 0.275% of contacts are cross-species, and none of them carries a transfer, so the model is evaluated almost entirely on within-species pairs. The held-out split is taken over snapshot pairs rather than over simulation runs, so it measures generalisation to unseen time windows of runs seen in training, not to unseen simulations. Three of the eleven modelled genes are moved by a mechanism that is not their real one, and several parameters that resemble measurements are invented; all are labelled as such. No external validation against real genomic data exists: our attempt failed because the isolate data we could obtain carries resistance phenotypes rather than per-isolate gene calls. Under these conditions we read a high AUROC as a statement about the difficulty of the task as posed rather than as evidence that the model has learned conjugation. What we take to be transferable is the formulation — an exactly labelled, per-contact, per-gene transfer task — and the finding that training one model across all genes reaches rare targets that per-gene models cannot.

1 Introduction

Antimicrobial resistance (AMR) is a major and growing threat to the treatment of bacterial infections [1, 2]. A central route by which resistance spreads between bacteria is horizontal gene transfer (HGT), in which resistance genes move from one cell to another — for example by plasmid conjugation — rather than arising independently in each lineage [3, 4]. Whether a transfer happens depends on local, cell-level conditions such as contact between cells and the physiological state of the donor [5, 6]. In natural bacterial communities, however, there is a gap between what is known about HGT from laboratory experiments and what is known from natural environments [7], and even reliably assigning mobile genetic elements to their host cells in such communities has been difficult [8].

Agent-based simulation offers a setting in which every transfer event is recorded. In an agent-based (individual-based) model, each bacterium is an explicit agent that follows stated rules [9], and such models have been used to study plasmid transfer in spatially structured populations [10, 6]. In our model these rules cover growth, death, stress responses and gene exchange, so the ground truth of *which cell passed which gene to which neighbour* is known exactly. This makes it possible to pose a supervised learning problem that is not available from observational data: given a snapshot of a bacterial population, predict which resistance genes will be transferred along which cell–cell contacts in the next time window. Graph neural networks (GNNs) are a natural fit for this problem, because the population can be represented as a graph of cells (nodes) connected by spatial proximity (edges), and the prediction target is defined per edge; we use graph attention networks [11].

In this paper we build such a system and evaluate it with the aim of reporting only what the numbers support. We develop an agent-based AMR simulation with explicit conjugation-style HGT, and a graph attention network that predicts per-gene transfer on each directed cell–cell edge. All reported results are within-simulation: the labels are the simulator's own recorded transfer events, and we have not yet validated the predictions against real genomic data (Section 5).

Scope. Our simulator models *intraspecies* transfer only: a gene moves between two cells of the same species, so a contact between cells of different species is a negative example by construction. This is a substantial restriction rather than an incidental one, because interspecies and intergenus transfer of several of the exact genes we model is documented — the *mcr-1* colistin-resistance plasmid was conjugated into *Escherichia coli* and maintained in *Klebsiella pneumoniae* and *Pseudomonas aeruginosa* [12], and an NDM-1 carbapenemase plasmid was conjugated from *Citrobacter freundii* into an *E. coli* recipient [13]. The prediction task we report on is therefore easier than the corresponding task in a community of mixed species. A second qualification applies to the evaluation itself: our held-out split is taken over snapshot pairs rather than over simulation runs, so it measures generalisation to new time windows within runs seen during training, and not to unseen runs, scenarios or species. Both restrictions should be kept in mind when reading the figures below (Sections 3, 5 and 6).

Our contributions are:

1. **A per-edge, per-gene HGT prediction task grounded in recorded simulation events**, with a documented simulation whose biology exists in two explicitly versioned forms. Results are reported on the corrected version (Section 3).
2. **A GNN that outperforms a per-gene Random Forest baseline, under the evaluation described above.** On the four resistance genes our per-gene Random Forest baseline can be trained on in every seed, the GNN reaches a test AUROC of 0.9777 ± 0.0023 , against 0.9548 ± 0.0124 for the Random Forest, and is better on all 5 of 5 seeds (Section 4). Both models are evaluated

on the same held-out snapshot pairs, so this is a like-for-like comparison between the two approaches on this task; it is not evidence about either model's behaviour on unseen runs or on interspecies transfer, neither of which we test.

3. **Coverage of rare genes.** Our per-gene Random Forest and logistic-regression baselines, as implemented, need at least five positive examples in the training subsample they draw, and that subsample is drawn per seed. Two genes fall below the threshold in every seed (*acrAB-tolC* and *gyrA_S83L*, with 19 and 34 positive transfer events in the whole dataset), and a third falls below it in three seeds of five (*blaKPC-2*, with 94). The GNN, trained jointly on all genes, gives stable predictions for all three (Section 4). We note which is which rather than treating them alike, and we flag that *acrAB-tolC* and *gyrA_S83L* are chromosomal in real bacteria and are moved between neighbouring cells by our simulator as a deliberate simplification, so their transfer events are not models of documented conjugative transfer (Sections 3 and 5); *blaKPC-2* carries no such caveat.
4. **An honest account of where the signal comes from.** Message passing over the contact graph adds a small but consistent gain ($+0.0031 \pm 0.0014$ AUROC, better on 5 of 5 seeds). The genes a cell already carries are by far the most important input, while edge features and local antibiotic exposure add no measurable signal in our setting. For antibiotic exposure this is under our dosing protocol, in which drug is present for only 5 of the 80 simulated steps (Section 4).
5. **A reproducible pipeline:** the simulation and training-data generation are seeded and deterministic across processes (verified byte-for-byte for the frozen `paper_v1` configuration), GPU training with the same seed reproduces test AUROC to within 2.1×10^{-4} , and every reported number is a mean $\pm$ standard deviation over independent model seeds (Section 3).

2 Related Work

2.1 Individual-based models of bacteria and plasmid transfer

Individual-based (agent-based) models represent each microbial cell explicitly. They have been used to study how the behaviour of individual cells gives rise to population-level patterns, from biofilms to phage–CRISPR dynamics [9]. General-purpose frameworks include iDynoMiCS, for individual-based modelling of biofilms [14], and BSim, an agent-based tool for bacterial populations in systems and synthetic biology [15].

Several studies model plasmid transfer specifically. - Krone et al. present an individual-based lattice model of plasmid transfer and persistence in spatially structured populations, and compare its predictions with agar-surface experiments [10]. - Merkey et al. extend an individual-based biofilm model with plasmid carriage and transfer by individual cells. They use it to show, *in silico*, that growth-dependent conjugation limits plasmid invasion of biofilms [6]. - Seoane et al. use an individual-based experimental framework to estimate the main parameters governing conjugation at the single-cell scale [5]. - At a much coarser scale, the VERA model simulates the spread of resistance between people, including transfer of resistance determinants from commensal gut bacteria to a pathogen [16].

These models use simulation to test mechanistic hypotheses about how plasmids and resistance spread. We use simulation for a different purpose: as a source of exactly labelled transfer events for training and evaluating a predictive model. This purpose carries a limitation that the work above does not share to the same degree. Krone et al., for example, compared their model against experiments. Our simulator's

per-step transfer probabilities are not fitted to experimental conjugation data, and all of our results are within-simulation (Section 5).

2.2 Machine learning for resistance and gene-transfer prediction

Machine learning is increasingly used to predict antimicrobial resistance phenotypes from bacterial gene content and genome composition. A recent review notes that such models typically treat genes as independent predictors [17].

Closer to our task, two lines of work predict the transfer of resistance genes rather than the resistance phenotype. - Ellabaan et al. identify putative horizontally transferred resistance genes and the gene-exchange networks that disseminate them across bacterial genomes. They then use the associated mobilisation elements to forecast where known resistance genes may spread [18]. The confirmatory analysis in that paper was later revised in an Author Correction. - Zhou et al. predict a network of recent HGT events between genomes from their functional gene content [19]. An event is defined by near-identical DNA shared by distantly related organisms. A random forest on functional profiles gave the best performance. A baseline graph convolutional network performed similarly to it, and improved as network-topology information was added.

Both operate on real genomes. Transfers are therefore inferred from sequence similarity, at the level of genomes and over evolutionary time. Our task is defined at the level of individual cells and their contacts, over the next simulated time window, with labels taken from recorded events. The two settings trade off in opposite directions. Genome-level work has real data but indirect labels; ours has exact labels but simulated data.

2.3 Graph neural networks for antimicrobial resistance

Graph neural networks have recently been applied to resistance problems in which graph nodes are isolates or patients. - AMR-GNN treats each bacterial isolate as a node, connects isolates by genomic distance, and predicts resistance phenotypes by node classification [20]. - Donabauer et al. apply graph neural networks to time-dependent graphs of patient movements within a hospital to identify carriers of vancomycin-resistant enterococci [21]. - The graph convolutional network of Zhou et al. (Section 2.2) operates on a network of genomes [19].

Our graph is at a finer scale. Nodes are individual simulated bacterial cells, edges are spatial contacts between them, and the prediction target is defined per edge and per gene: whether a given resistance gene is transferred along a given contact. We use graph attention [11] for this.

In the literature we reviewed, we did not find prior work that poses per-contact, per-gene HGT prediction among individual cells as a supervised learning problem. We position our contribution as bringing together the individual-based modelling of plasmid transfer (Section 2.1) and graph-based learning for resistance (Sections 2.2–2.3). We do not claim to identify a previously unstudied problem.

3 Methods

3.1 Agent-based simulation

The simulator is an agent-based model built on Mesa 3.5.1. Bacteria occupy an 80×60 multi-agent grid on which several cells may share a position. Each simulated step, every living cell updates its energy, its

stress response and its persister state, takes antibiotic damage, attempts mutation, attempts gene transfer, updates its biofilm status, moves and attempts division, in that order.

Five bacterial species are modelled (*Escherichia coli*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa* and methicillin-resistant *Staphylococcus aureus*), each with its own doubling time, baseline fitness, baseline mutation rate, biofilm propensity, motility, intrinsic resistance genes and a set of genes it may acquire. Ten resistance genes are modelled, drawn from CARD, spanning β -lactamases (blaTEM-1, blaCTX-M-15, blaKPC-2, blaNDM-1), efflux systems (mexAB-oprM, acrAB-tolC), target alteration (gyrA_S83L, mcr-1), ribosomal protection (tetM) and glycopeptide resistance (vanA); the corrected biology adds mecA as an eleventh. Six antibiotics are modelled (ciprofloxacin, meropenem, colistin, vancomycin, ampicillin, tetracycline), each as a concentration field over the grid that diffuses and decays.

3.1.1 Antibiotic action

A cell's protection against a drug is combined multiplicatively over its genes, as independent probabilities, and is then raised by biofilm membership and by the persister state. For a bactericidal drug, damage accrues each step as a sigmoid (Hill-type) function of the locally effective concentration,

$$\text{damage} = E_{\text{max}} \cdot C^n / (EC_{50}^n + C^n),$$

with EC₅₀ set to the drug's minimum bactericidal concentration and, in the code, $E_{\text{max}} = 0.3$ and $n = 2$. Damage accumulates; a cell begins dying at 0.7 and dies at 1.0. Bacteriostatic drugs delay division instead of causing damage. **E_{max} and the Hill coefficient are invented parameters:** neither is taken from a cited source, and because no real duration is assigned to a simulated step, neither the damage rate nor the per-drug decay rates can be compared against measured pharmacokinetics. We report them as simulator settings, not as pharmacological estimates.

3.1.2 Horizontal gene transfer, and the four ways a cell gains a gene

Transfer is modelled as conjugation between neighbouring cells. A cell attempts transfer on a randomly chosen 30% of its steps. When it does, it considers every cell in its immediate (Moore radius 1) neighbourhood and, for each gene it carries, transfers that gene with probability

$$p = \text{acquisition_prob}(\text{gene}) \times (2 \text{ if the donor's SOS response is active}) \times m_{\text{biofilm}},$$

where m_{biofilm} is 4 when both cells are in a biofilm, 1.5 when one is, and 1 otherwise, and p is capped at 0.95. The recipient must not already carry the gene, and the gene must be in the recipient species' acquirable set. The biofilm multipliers follow the observation that biofilm matrices raise conjugation frequency; **the per-gene acquisition_prob values (0.01–0.05 per step) and the 30% attempt rate are invented parameters with no cited source.**

Three restrictions define the task, and all three must be read alongside the results:

1. **Transfer is intraspecies only.** A donor skips any neighbour of a different species, so every cross-species contact is a negative example by construction. Real transfer of several of these exact genes is documented across species and genera: the mcr-1 plasmid was conjugated into *E. coli* and maintained in *K. pneumoniae* and *P. aeruginosa* [12]; an NDM-1 plasmid was conjugated from *Citrobacter freundii* into an *E. coli* recipient [13]; Tn916-borne tet(M) transferred from *Lactococcus lactis* to *Enterococcus faecalis* in vitro and in a rat gut [22]. The task we evaluate is therefore easier than transfer prediction in a mixed-species community.

2. **Two genes are moved by a mechanism that is not their real one.** *gyrA_S83L* is a chromosomal target-site mutation, and quinolone resistance that is plasmid-borne is carried by a different set of genes (*qnr* proteins, a modifying enzyme, mobile efflux pumps) rather than by mobilised *gyrA* [23, 24]. *acrAB-tolC* is a chromosomal RND efflux system, ubiquitous in the Gram-negative species that carry it, and resistance arises by overexpression rather than by gaining the genes [25]. The simulator nonetheless transfers both between neighbours. We label their transfer events a **simplified mechanism** and do not present them as models of conjugative transfer. (Plasmid-borne RND pump genes do exist [25]; what the model does that reality does not is move a species' own resident *AcrAB-TolC* between cells of that species.) *vanA* is simplified in a different way: it arises and spreads within MRSA, whereas the documented route is interspecies transfer of Tn1546 from *E. faecalis* [26, 27], and *Enterococcus* is not modelled. *vanA* is reported separately throughout. For *gyrA_S83L* we use the generic description "a *gyrA* target-site mutation": the gene object carries CARD's *E. coli gyrA* entry but is available to three species in the model, and we did not establish that the S83L substitution specifically is the dominant variant in all three, so the particular variant modelled should be treated as unverified.
3. **Two genes are never transferred, correctly.** *mexAB-oprM* appears only as an intrinsic gene and is in no species' acquirable set, so it has no transfer events by construction and is excluded from evaluation. *mecA* is marked non-transferable because its real mobilisation is by SCCmec, a mechanism the model does not implement; reusing conjugation for it would be the wrong mechanism.

Besides transfer, a cell can gain a gene by inheriting it at division, by starting with its species' intrinsic genes, or through a mutation step. The mutation step has a branch that adds a **uniformly random gene from the species' acquirable set with no donor at all**. This last route has no biological counterpart — an acquired gene must come from somewhere — and exists only to seed genes into the population. It matters for interpreting the results, because no acquired gene is present at initialisation, so every gene's first carrier arises this way. How readily a gene seeds therefore depends on the size of its species' acquirable set, on whether it has an additional seeding route, and on how many scenarios contain a species that can carry it; and however it seeds, the resulting count is then amplified by spread, so that the earlier a gene happens to seed, the more events it accumulates. **The number of transfer events per gene is consequently an outcome of the seeding code and of chance rather than of gene epidemiology.** Consistent with that, three of the genes in our dataset share the same species availability and near-identical transfer probabilities yet differ roughly two-fold in their event counts. We therefore do not interpret per-gene differences biologically anywhere in this paper.

Only recorded transfer events produce the supervised labels; genes gained by the other three routes produce no label.

3.2 Two versioned biologies

The simulation's biology exists in two explicitly versioned forms, and we report on the corrected one.

`paper_v1` is the original configuration, frozen. A regression test checks that its simulated states and its generated training pairs are byte-identical to the pre-versioning code under more than one Python hash seed.

`lab_v2` is the corrected configuration and the one all reported results use. Its differences from `paper_v1`, each with the source that motivated it, are:

Change	Basis
MRSA's intrinsic gene is <i>mecA</i> rather than <i>tetM</i>	CARD ARO:3000617 describes <i>mecA</i> (PBP2a) as commonly associated with MRSA; CARD ARO:3000186 describes <i>tet(M)</i> as found on transposable elements, and EUCAST Expected Resistant Phenotypes v1.2 does not list <i>S. aureus</i> as expected-resistant to tetracyclines
<i>acrAB-tolC</i> removed from MRSA's acquirable set	<i>AcrAB-TolC</i> is a Gram-negative tripartite system (CARD ARO:3000384)
<i>mecA</i> is non-transferable	SCCmec mobilisation is not modelled (§3.1.2)
<i>K. pneumoniae</i> 's blanket intrinsic <i>acrAB-tolC</i> replaced by species-level intrinsic resistance to ampicillin only	EUCAST Expected Resistant Phenotypes v1.2 rule 1.7 lists the <i>K. pneumoniae</i> complex as expected resistant to ampicillin/amoxicillin and ticarcillin, not to ciprofloxacin or tetracycline; ticarcillin is not simulated
Antibiotic diffusion conserves total drug	Bug fix: the original applied a kernel that already summed to one and then rescaled the field, removing drug every step, which contradicts a diffusion process
<i>mecA</i> fitness cost 0.275	Ender et al. 2004, comparing the strains RA120 and BB255

A known limitation carried by both versions is that decay rates are unvalidated against real pharmacokinetics, for the reason given in §3.1.1.

We also note two residual issues we did not change, because changing them would alter the dataset: *A. baumannii* carries *mexAB-oprM* and *acrAB-tolC* as intrinsic genes although its characterised RND systems are AdeABC, AdeIJK and AdeFGH [28]; and *blaTEM-1* is available only to *E. coli*, although it is common in *K. pneumoniae* [29].

3.3 Prediction task and dataset

3.3.1 Graphs and labels

From a simulation snapshot we build a graph whose nodes are living cells (subsampled to at most 300 per snapshot with a seeded generator) and whose edges join pairs of cells within three grid cells of one another, added in both directions. Each node carries a 35-dimensional feature vector: ten binary gene-presence flags; five physiological values (fitness, energy, stress level, accumulated antibiotic damage, normalised age); two behavioural flags (biofilm membership, persister state); normalised position; three population descriptors (local density, generation, offspring count); a five-way species one-hot; a two-way Gram stain indicator; and the local concentration of each of the six drugs. Each edge carries five features: normalised shared-gene count, absolute fitness difference, a both-in-biofilm flag, absolute stress-level difference, and a coarsened proximity band with three levels (same cell / within three cells / farther).

The edge radius of three is deliberately wider than the transfer radius of one. If the graph contained only transfer-eligible pairs, the existence of an edge would itself identify eligibility, which would put the label-generating gate back into the graph's topology. Including near-but-ineligible pairs under the same coarse proximity band prevents that.

The target is defined per directed edge and per gene: did a recorded transfer of that gene occur from the source cell to the target cell within the snapshot window? Labels come from the simulator's own event log, recorded at the moment a transfer fires, and never from comparing genomes between snapshots. This distinction is necessary because a cell can gain a gene by mutation in the same window in which an

eligible neighbour happens to carry it; inferring transfer from genome differences would label that as a transfer from a cell that had nothing to do with it.

3.3.2 Data generation

Each run initialises 120 cells, steps 15 times, then applies its scenario's drugs uniformly and continues to 80 steps, taking a snapshot every three steps and pairing consecutive snapshots. Under the corrected biology the dose is 0.25 µg/mL, cleared after five further steps; this time-limited course is an invented protocol, expressed in steps because no real step duration is defined. Five scenarios are used — `ecoli_cipro`, `klebsiella_carbapenem`, `pakistan_crisis`, `xdr_acinetobacter` and `mrsa_hospital` — each with three data seeds.

The resulting dataset (SHA-256 `ee83ff385ca15ec9...`) contains 390 graph pairs, 1,707,498 edges and 1,443 positive transfer events. Positives are distributed very unevenly across genes — `blaCTX-M-15` 480, `tetM` 420, `mcr-1` 179, `blaNDM-1` 177, `blaKPC-2` 94, `vanA` 35, `gyrA_S83L` 34, `acrAB-tolC` 19, `blaTEM-1` 5 and `mexAB-oprM` 0 — for the structural reason given in §3.1.2, not for any biological reason. Against ten gene outputs the positive rate is approximately 8.5×10^{-5} per edge-gene pair.

3.3.3 Splitting, and a caveat about it

Graph pairs are shuffled with a fixed seed and split 70% / 15% / 15% into training, validation and test sets. The same split seed is used by the neural model, by the baselines and by calibration, so that all of them are evaluated on the same held-out edges; this was not originally the case, and correcting it is part of the history in §3.4.

The split is over graph pairs, not over simulation runs. Because each run contributes roughly 26 pairs taken three steps apart, snapshots from one run — sharing its founding population, its seed and largely the same cells — can be distributed across training, validation and test. The held-out set therefore measures generalisation to new time windows of runs the model has seen, not to unseen runs, scenarios or species. This inflates the reported figures relative to a grouped split, and we state it here rather than leaving it to be inferred (see also Section 5, and Section 7.1 for what fixing it would involve).

3.4 Development history that affects the numbers

Two corrections separate the numbers reported here from earlier ones, and we record them because they change what the figures mean rather than merely improving them.

Label definition. Labels were originally inferred by comparing genomes between snapshots, which conflates transfer with independent mutation and produces positives attributable to a cell that did not donate anything. Labels now come from recorded events only.

Feature leakage. Four inputs were direct components of the rule that generates the label: the same-species edge flag and the raw edge distance (both hard preconditions for transfer), a transferable-gene count that mirrored the eligibility test almost exactly, and the SOS flag (the literal term that doubles the transfer probability). All four were removed. Node features went from 36 to 35 dimensions and edge features from 8 to 5. Raw distance was replaced by the coarsened proximity band described in §3.3.1, and the transferable-gene count by a plain shared-gene count. We confirmed which layout produced the earlier results by rebuilding the architecture under both: the pre-correction layout reproduces a parameter count of 349,370 and the corrected layout 349,162, matching the counts logged by the respective runs. Results obtained before these two corrections are not comparable with those reported

here and are not reported. An audit file referenced in a source comment as holding the full leakage analysis is not present in the repository; §3.4 is the account of record.

Separately, results obtained with the original default hyperparameters are superseded by the tuned configuration of §3.5, selected on the validation split.

3.5 Model

The model is a graph attention network [11]. Node features are encoded by one small linear encoder per feature group — genomic, physiological, behavioural, spatial, population, species, Gram stain, drug exposure — whose outputs are concatenated and projected to the hidden width, so that groups with different scales and meanings are not mixed by a single projection. Edge features are encoded separately by a two-layer network.

Each message-passing block applies multi-head graph attention that takes the encoded edge vector as an edge attribute, followed by a residual connection and layer normalisation, then a position-wise feed-forward network with a further residual connection and normalisation. For each directed edge, a prediction head receives the source representation, the target representation and the encoded edge vector, and emits one logit per gene.

The configuration used for all reported results, selected on the validation split, is: hidden width 128, edge encoding width 64, two attention blocks, four heads, dropout 0.15. Training uses binary cross-entropy with a positive-class weight of 15 to offset the extreme imbalance, AdamW at learning rate 10^{-3} with weight decay 10^{-4} and cosine annealing, gradient clipping at 1.0, batch size 8, and at most 60 epochs. Early stopping watches validation AUROC with patience 12 and a minimum improvement of 10^{-3} , but is suppressed for the first 20 epochs.

That warm-up is not cosmetic. Without it, one seed stopped at epoch 13 because validation AUROC had peaked at epoch 1 — on an essentially untrained model — and had not yet recovered from a subsequent dip; the retained checkpoint was worse on every gene. The behaviour was reproducible on retraining, and the warm-up resolves it: that seed reaches the same range as the others, while a normally-behaved seed is unchanged. With the warm-up in place, every seed in the reported runs trained for 27 to 38 epochs with its best epoch between 15 and 32.

Edge features are zeroed in all reported runs. Tuned models with and without edge features were indistinguishable on the frozen biology, so the reported configuration does not use them. This should not be read as evidence that edge attributes are harmful; the one earlier result that appeared to show a cost was the early-stopping artefact described above.

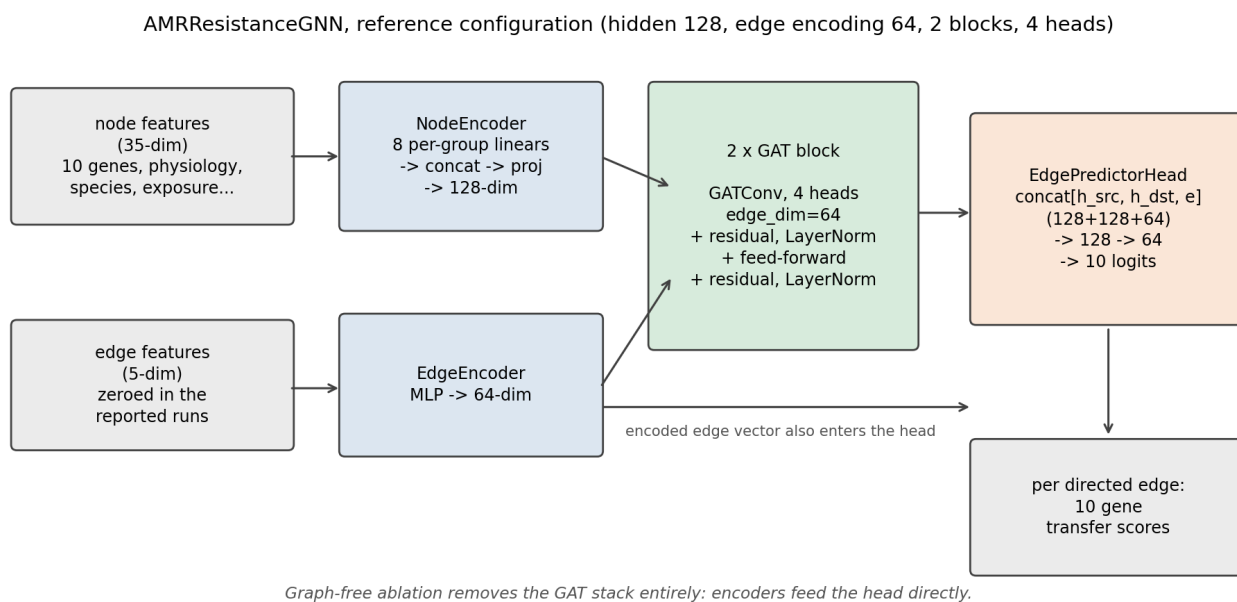


Figure 1. Model architecture, reference configuration. Schematic of AMRResistanceGNN as configured for every reported result: node features (None-dim) through a per-group NodeEncoder to width 128, edge features (None-dim) through an EdgeEncoder to width None, then 2 graph-attention blocks (None heads, each with a residual connection, layer normalisation and a feed-forward sublayer), then a head that concatenates the two endpoint representations with the encoded edge vector and emits None per-gene logits for each directed edge. Edge features are zeroed in all reported runs, so that pathway is present but carries no information. The graph-free ablation removes the attention stack, leaving the encoders feeding the head directly, and costs 0.0031 ± 0.0014 headline AUROC (claim S4). Configuration read from `gnn_hparams` and `feature_dims`; layer structure from `ai/gnn_model.py`.

3.6 Baselines and ablations

Per-gene classical baselines. Random forest and logistic regression are trained one-versus-rest per gene on the flattened concatenation of the two endpoint node vectors and the edge vector, using a seeded 100,000-edge subsample of the training edges. The random forest uses 300 trees, maximum depth 8 and balanced subsampling. A gene with fewer than five positives in that subsample cannot be trained and is scored 0.5 by construction. We keep this cutoff as it is: lowering it for the comparison would change the baseline rather than test it.

Because the subsample is drawn independently for each seed, **trainability is a per-seed property**: two genes fall below the threshold in every seed, and a third falls below it in three seeds of five. The headline comparison is therefore computed on the four genes the random forest fits in *every* seed, so that no 0.5-by-construction score enters it. The genes it fits unreliably or never are reported separately under coverage, as a property of the approaches rather than folded into the same number.

Message-passing ablation. The same architecture with the attention blocks removed, so that node and edge encodings feed the prediction head directly, isolates the contribution of message passing.

Feature-group ablation. Groups of node features are zeroed one at a time to measure each group's contribution.

3.7 Evaluation

Genes are handled according to a fixed, pre-recorded policy. *mexAB-oprM* is excluded, having no transfer events by construction. *vanA* is evaluated but always reported separately and labelled a simplified mechanism. The headline macro average is taken over the remaining eight genes. One of those eight, *blaTEM-1*, has no positives in the test split (five in the whole dataset), so no model can be scored on it and the macro average is effectively over seven; we say so wherever the figure appears rather than implying eight evaluable genes.

We report AUROC as the primary metric, since the positive rate makes accuracy uninformative, alongside AUPRC and expected calibration error. Every figure is a mean and standard deviation over five independent model seeds, with three seeds for the ablations; the seeds vary model initialisation and training, not the data. Per-gene AUROCs are reported as discrimination figures only, always with their positive counts, and are not interpreted as reflecting the relative mobility of real genes, for the reason given in §3.1.2.

Data generation and training are seeded and deterministic across processes: identical simulated states and identical training pairs are produced under different Python hash seeds, which a regression test enforces for the frozen biology. Retraining the same seed on a GPU reproduces test AUROC to within 2.1×10^{-4} .

4 Results

All figures are means and standard deviations over five independent model seeds (three for the ablations), on the held-out split described in Section 3.3.3. That split is over snapshot pairs rather than over runs, so throughout this section "test" means unseen time windows of runs that contributed training data, not unseen runs, scenarios or species. Unless stated otherwise, "headline AUROC" is the macro average over the eight genes fixed by the evaluation policy of Section 3.7, one of which (*blaTEM-1*) has no positives in the test split, so the macro is in practice over seven.

4.1 The GNN outperforms per-gene classical baselines

On the four genes the per-gene random forest can be trained on **in every seed** — *blaCTX-M-15*, *blaNDM-1*, *mcr-1* and *tetM* — the GNN reaches a test AUROC of **0.9777 ± 0.0023** against **0.9548 ± 0.0124** for the random forest. The GNN is ahead on **all five of five seeds**. We treat this as the primary comparison, because it is the only one in which both models are actually fitted to the same genes in every run.

The qualifier "in every seed" is doing real work here, and an earlier version of this analysis got it wrong. The baselines draw their 100,000-edge training subsample independently per seed, so a gene's trainability is a per-seed property rather than a fixed one. *blaKPC-2*, with 94 positive events dataset-wide, clears the five-positive threshold in only two of the five seeds; in the other three the random forest scores it 0.5 by construction. Including it in this comparison — as we initially did — folds that construction artefact into the baseline's score, which is precisely what §4.1's closing paragraph says must not be done. It inflated the gap roughly threefold, from 0.023 to 0.079, and the tell was the baseline's implausibly large standard deviation, which turned out to be bimodal across seeds. *blaKPC-2* is reported under coverage in §4.2 instead.

The GNN's own headline macro, over the eight genes of the evaluation policy, is **0.9805 ± 0.0012**. Logistic regression reaches **0.7096 ± 0.0359** on the same headline macro.

We deliberately do not quote a headline-macro *difference* between the GNN and the random forest. The random forest's headline macro is depressed by the genes it cannot reliably train, which are scored 0.5 by construction rather than by any prediction; a difference computed across all eight genes would therefore combine a genuine performance gap with a coverage gap and overstate the former. The two are reported separately: performance in this subsection, on the four genes both models always fit, and coverage in §4.2.

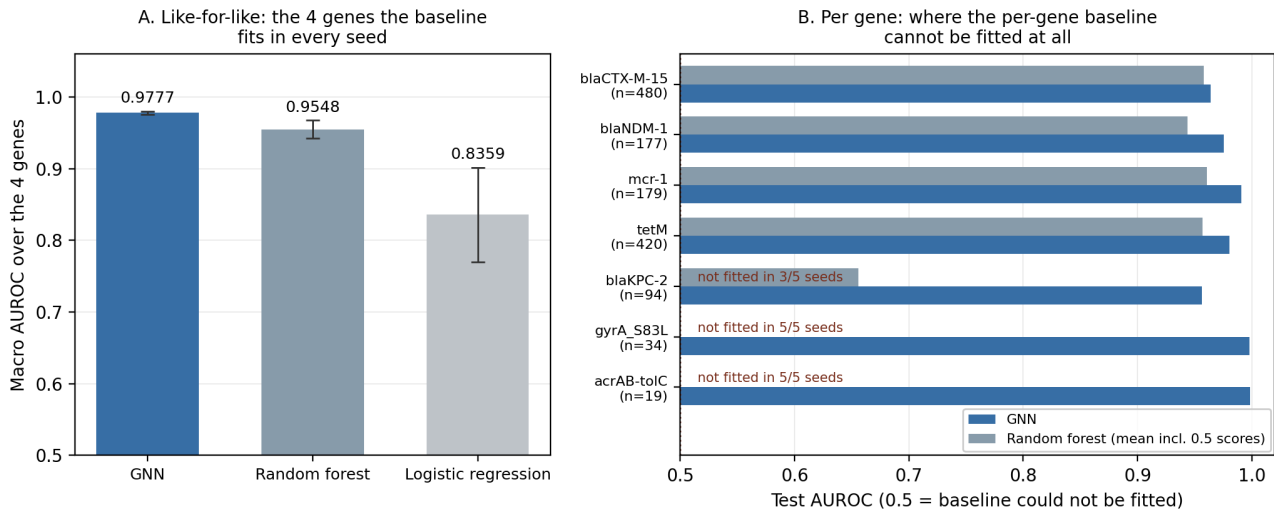


Figure 2. Headline comparison, and where the per-gene baseline cannot be built. Panel A. Macro AUROC over the four genes the random forest fits in *every* seed (blaCTX-M-15, blaNDM-1, mcr-1, tetM): GNN 0.9777 ± 0.0023 , random forest 0.9548 ± 0.0124 , logistic regression 0.8359 ± 0.0658 ; the GNN is ahead on 5 of 5 seeds (claim S2). This set deliberately excludes blaKPC-2, because the baselines draw their 100,000-edge training subsample per seed and blaKPC-2 clears the five-positive threshold in only 2 of 5 seeds; including it would fold 0.5-by-construction scores into the baseline, which is the artefact claim U5 exists to keep out of a performance comparison. An earlier version of this analysis did include it and overstated the gap roughly threefold (0.079 against the correct 0.023). **Panel B.** Per-gene view of the same run, ordered as in the text, with each gene's total positive count. The dotted line marks 0.5, the score assigned when a gene cannot be fitted at all. acrAB-tolC (19 positives) and gyrA_S83L (34) are never fitted in any seed; blaKPC-2 (94) is fitted in 2 of 5, and the plotted random-forest bar for those three is a mean that includes those 0.5 scores, so it is a summary of availability rather than of skill. The GNN produces a stable prediction for every gene shown in every seed. Claims S2, S2b, U5. Values from `per_seed[*].per_gene_auroc` and `dataset.positives_per_gene`.

4.2 Coverage of genes the baselines cannot reliably train on

Our per-gene random forest and logistic-regression baselines require at least five positive examples for a gene within their 100,000-edge training subsample, and that subsample is drawn independently for each seed. Trainability is therefore a per-seed property, and it comes in degrees. The GNN, trained jointly on all ten gene outputs, produces stable predictions across all five seeds for every gene below:

Gene	Positives	Baselines trainable	GNN test AUROC	Random forest
acrAB-tolC*	19	0 of 5 seeds	0.9986 ± 0.0018	0.5 by construction, always
gyrA_S83L*	34	0 of 5 seeds	0.9977 ± 0.0006	0.5 by construction, always
blaKPC-2	94	2 of 5 seeds	0.9564 ± 0.0021	0.9611 and 0.8179 where it trains

* transfer modelled by a simplified mechanism (Section 3.1.2).

We state the tiers separately because they are not the same claim. For *acrAB-tolC* and *gyrA_S83L* the baseline is simply unavailable. For *blaKPC-2* it is *unreliable*: it fits in two seeds out of five, and where it does fit it returns 0.9611 and 0.8179 — a spread of 0.14 on the same gene and the same data, against the GNN's 0.9564 ± 0.0021 . The practical difference between a model that sometimes cannot be built and one that is built but varies this much is smaller than it looks, and *blaKPC-2* is the cleaner illustration of the point, because unlike the other two it carries no mechanism caveat: it is a genuine plasmid-borne gene, mobilised by Tn4401 in reality as well as in our simulator.

Three qualifications belong with these figures, and none is incidental.

First, the counts. Nineteen, thirty-four and ninety-four positive events across the whole dataset are few, and the standard deviations above are over model seeds on a fixed split, so they do not reflect the sampling uncertainty that such counts imply. These AUROCs should not be read as precise.

Second, the mechanism, for two of the three. *acrAB-tolC* and *gyrA_S83L* are chromosomal in real bacteria and are moved between neighbouring cells by the simulator as a deliberate simplification (Section 3.1.2), so their transfer events are not models of documented conjugative transfer. What their result demonstrates is a property of the learning setup — that joint training across genes yields usable predictions for targets too rare to fit individually — and not a biological finding about those genes. *blaKPC-2* carries no such caveat.

Third, this is a claim about our baselines as implemented, not about random forests. A different baseline design — a single multi-label forest, or a lower positive threshold — would face this differently, and we did not lower the threshold precisely because doing so would change the baseline rather than test it.

With that scope stated, the coverage difference is a real advantage of the approach rather than an artefact of the comparison: it follows from training one model on all genes instead of one model per gene. It is a property of the baselines *as we implemented them*, namely one-versus-rest with a five-positive requirement, and not of random forests in general. We did not lower that threshold, since doing so would have changed the baseline rather than tested it.

The GNN is evaluated on seven of the eight policy genes plus *vanA* separately. The eighth, *blaTEM-1*, has five positive events in the entire dataset and none in the test split, so no model of any kind can be scored on it; we report it as not evaluable rather than as a result.

4.3 Where the signal comes from

Message passing contributes a small but consistent gain. Removing the attention blocks, so that node and edge encodings feed the prediction head directly, costs **0.0031 ± 0.0014** headline AUROC (0.9805 with message passing against 0.9774 ± 0.0006 without). The direction is consistent across all five of five seeds. The effect is reliable and small: the graph-free model retains almost all of the performance, so most of what the model exploits is available from the two endpoint feature vectors alone rather than from the surrounding contact structure.

The genes a cell already carries dominate. Zeroing feature groups one at a time, only the genomic group matters appreciably: removing it costs 0.0549 ± 0.0331 headline AUROC (three seeds). Every other group changes the headline by less than ± 0.003 , which is within seed-to-seed variation and which we therefore do not interpret as an effect in either direction.

Local antibiotic exposure adds no measurable signal. Zeroing the six drug-concentration features changes the headline by $+0.0001 \pm 0.0001$ — that is, removing them does not hurt. This deserves an explicit caveat rather than a conclusion about antibiotic pressure: under our dosing protocol the drug is present for only five of the eighty simulated steps, so most snapshots carry no exposure to measure. The correct reading is that these features carry no signal *in this protocol*, not that antibiotic exposure is irrelevant to gene transfer.

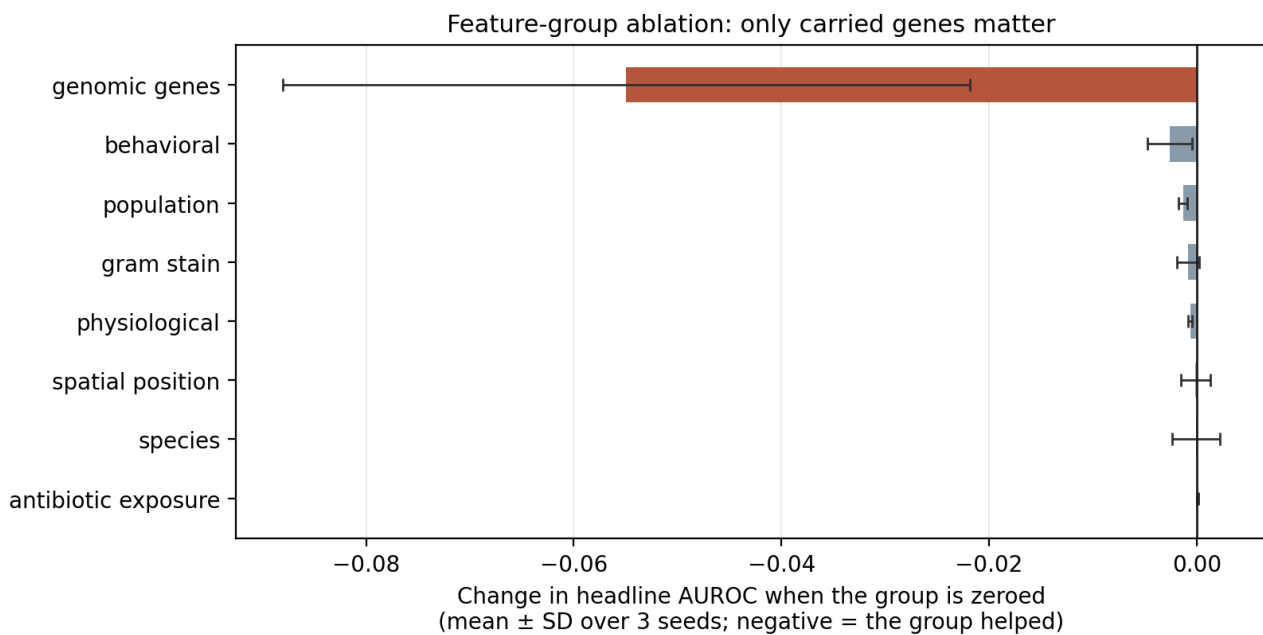


Figure 3. Feature-group ablation. Change in headline AUROC when each node-feature group is zeroed, mean $\pm$ SD over 3 seeds; more negative means the group contributed more. Only the carried-gene group matters: -0.0549 ± 0.0331 . Every other group falls within ± 0.003 , which is inside seed-to-seed variation and is not interpreted as an effect in either direction — including antibiotic exposure, for which the drug is present in only 5 of the 80 simulated steps under this dosing protocol (claim U4). Full-model reference: 0.9807. Values from `ablation`. Claims S5, U4.

4.4 Calibration and precision–recall

Predicted probabilities are well calibrated, with an expected calibration error of 0.0012 ± 0.0003 .

Headline AUPRC is 0.0316 ± 0.0293 . This is far above the base rate of approximately 8.5×10^{-5} positives per edge-gene pair, so the ranking carries real information in absolute terms; but the standard deviation is nearly as large as the mean, so the figure is unstable across seeds and we do not treat it as a headline result. The instability is expected given how few positives the rarer genes contribute.

4.5 Per-gene discrimination

Gene	GNN test AUROC	Positive events in dataset
acrAB-tolC	0.9986 ± 0.0018	19 — simplified mechanism
gyrA_S83L	0.9977 ± 0.0006	34 — simplified mechanism
mcr-1	0.9908 ± 0.0011	179
tetM	0.9805 ± 0.0033	420
blaNDM-1	0.9755 ± 0.0039	177
blaCTX-M-15	0.9642 ± 0.0030	480
blaKPC-2	0.9564 ± 0.0021	94
blaTEM-1	not evaluable	5, none in the test split

We offer no biological interpretation of the ordering in this table, because the variation in it is confounded. Two confounds account for it without any appeal to gene biology.

The first is **sample size, and it runs opposite to the direction that would make the table interpretable**. The positive counts span from 19 to 480, and the two highest AUROCs in the table belong to the two genes with the fewest positives (acrAB-tolC, 19; gyrA_S83L, 34), while the gene with by far the most positives (blaCTX-M-15, 480) sits near the bottom. A per-gene AUROC estimated from 19 positives is not comparable to one estimated from 480, and reading the ordering as a difficulty ranking would inverse-rank the genes by how much evidence supports each estimate.

The second is the **seeding artefact** of Section 3.1.2. No acquired gene exists at initialisation, so each gene's count is set by how readily the donor-free mutation branch seeds it and by how early that happens, after which spread amplifies the result. These counts are not a monotonic function of the per-step transfer probabilities: blaKPC-2, blaNDM-1 and mcr-1 are available to the same species with probabilities of 0.02, 0.015 and 0.02, yet record 94, 177 and 179 events. A roughly two-fold spread among otherwise matched genes is left unexplained by any parameter of the model, which is the clearest available indication that these counts are high-variance outcomes of the seeding process rather than stable properties of each gene.

A gene's position in this table therefore reflects the simulator's seeding code, the timing of a chance event, and the amount of evidence behind its estimate — not the mobility of the corresponding real gene. We report the values for completeness and for comparison against future versions of the pipeline, and draw no per-gene conclusions from them.

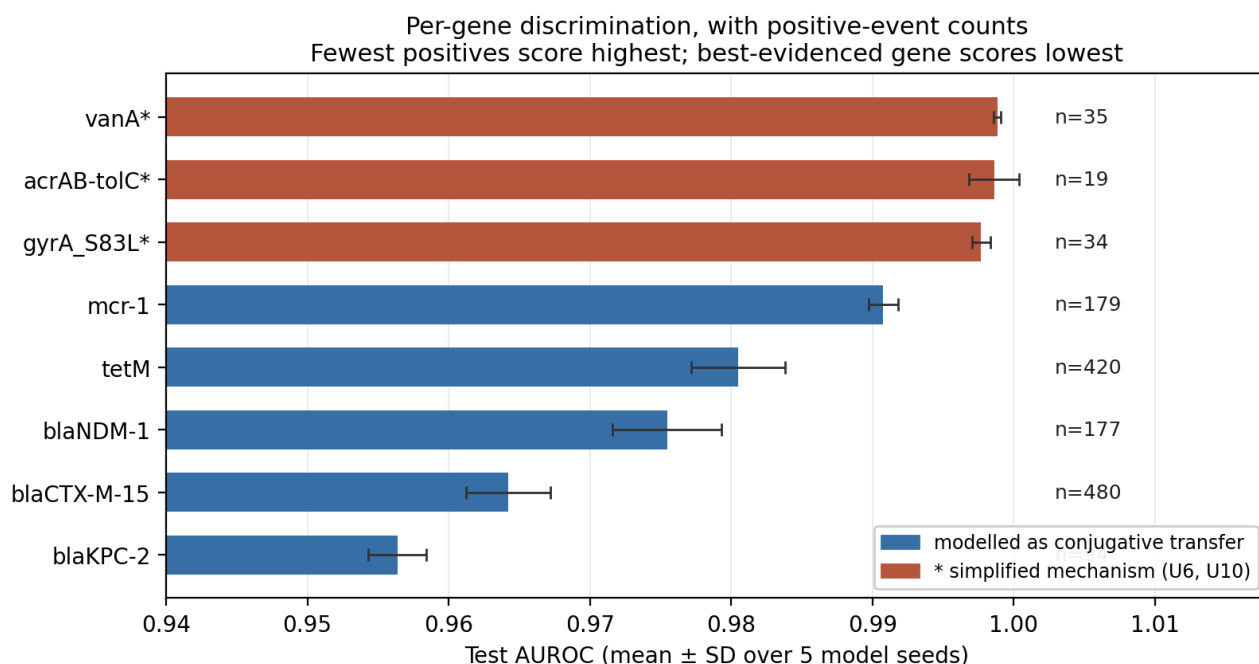


Figure 4. Per-gene discrimination, with positive-event counts. Per-gene test AUROC of the GNN, mean ± SD over 5 model seeds, each bar annotated with that gene's total positive transfer events in the dataset. **The ordering is confounded, and runs opposite to the evidence behind it: the two highest-scoring genes are the two with the fewest positives — *acrAB-tolC* 0.9986 from 19 events and *gyrA_S83L* 0.9977 from 34 — while the best-evidenced gene, *blaCTX-M-15* with 480 events, scores lowest at 0.9642.** An AUROC estimated from 19 positives is not comparable to one estimated from 480, so this figure must not be read as a difficulty ranking over genes. Starred genes (*acrAB-tolC*, *gyrA_S83L*) are chromosomal in real bacteria and are transferred by the simulator as a labelled simplification (claim U10); *vanA* is likewise simplified (U6) and is excluded from the headline macro. *blaTEM-1* (5 positives, none in the test split) and *mexAB-oprM* (0 positives, in no species' acquirable set) are not evaluable and are absent from the figure. Counts from `dataset.positives_per_gene`; AUROCs from `per_gene_auroc.gnn`. Claims S9, U6, U8, U10, U12.

4.6 *vanA*, reported separately

vanA reaches a test AUROC of 0.9988 on 35 positive events. It is excluded from the headline macro and reported here on its own, because the mechanism the simulator gives it is a simplification in two respects: the gene arises and spreads within MRSA, whereas the documented route is interspecies transfer of Tn1546 from *Enterococcus faecalis*, and *Enterococcus* is not modelled at all. This figure is a discrimination measurement on simulated events, not evidence about vancomycin resistance transfer.

mexAB-oprM is excluded entirely: it is present only as an intrinsic gene, appears in no species' acquirable set, and therefore has zero transfer events by construction. There is nothing to predict and no score to report.

4.7 Reproducibility

Data generation and training are deterministic across processes. Identical simulated states and identical training pairs are produced under different Python hash seeds, which a regression test enforces byte-for-byte for the frozen `paper_v1` configuration. Retraining the same seed on a GPU reproduces test AUROC to within 2.1×10^{-4} . Every figure in this section is a mean and standard deviation over

independent model seeds, with the data held fixed, so the reported spreads capture variation from initialisation and training order and not from data generation.

4.8 Summary

The GNN discriminates simulated per-edge, per-gene transfer events well (headline AUROC 0.9805 ± 0.0012) and beats a per-gene random forest on the four genes that baseline fits in every seed (0.9777 ± 0.0023 against 0.9548 ± 0.0124 , on five of five seeds). Its advantage in coverage — stable predictions for genes the per-gene baselines fit unreliably or not at all — follows from joint training. Most of the signal is carried by the genes a cell already holds; message passing adds a small, consistent increment; edge features and local drug exposure add nothing measurable in this setting. What none of these figures establish is stated in Section 5: no external validation, a held-out split that does not test unseen runs, and an intraspecies-only transfer model.

5 Limitations

We group these by what they limit: the claims the numbers support (§5.1), the biology the simulator implements (§5.2), the parameters it uses (§5.3), and the history of the pipeline itself (§5.4). Several are consequences of decisions we would make differently in a future version, and we say so where that is the case rather than presenting each as an inherent constraint.

5.1 What the evaluation does and does not establish

No external validation. Every number we report is internal to the simulator: the labels are its own recorded transfer events, and the model is never tested against real genomic data. We attempted such a validation and it is not functional. The public isolate collections we obtained are antimicrobial-resistance *phenotype* tables with no per-isolate gene calls, so the comparison degenerates — gene prevalences collapse to 1.0 or 0.0 and the rank correlation is undefined. A meaningful comparison requires re-annotating assemblies with a resistance-gene caller to obtain the gene-level calls the model predicts over. We scope that as future work and make no claim of external validity here. Consequently nothing in this paper should be read as a clinical or epidemiological prediction.

The held-out split does not test unseen simulations. We split graph pairs, not simulation runs. Each run contributes roughly twenty-six snapshot pairs three steps apart, and those pairs are distributed across training, validation and test, so a test pair typically comes from a run whose other windows were trained on — sharing its founding population, its random seed and largely the same individual cells. The reported figures therefore measure generalisation to unseen *time windows within seen runs*, and not to unseen runs, unseen scenarios or unseen species. This inflates them relative to a grouped split. A by-run or by-scenario split is the correct design, and we did not adopt it; doing so would change every number in Section 4 and requires a full regeneration, which we identify as the single most important change for a future version.

The comparison is between our implementations, not between model families. Our per-gene random forest and logistic-regression baselines are one-versus-rest with a five-positive training requirement, applied to a subsample drawn independently per seed. Two genes fall below it in every seed and a third in three seeds of five; in those cases the baseline is scored 0.5 by construction. This is a property of how we implemented those baselines, not a limitation of random forests, and we do not

claim otherwise. We kept the threshold rather than lowering it, because lowering it would have produced a different baseline rather than a fairer test of this one — but the consequence is that the coverage difference in §4.2 speaks to a design choice we made about the baselines as much as to the GNN.

For the same reason we do not quote a macro-averaged difference between the GNN and the random forest across all eight policy genes. Such a difference would combine a performance gap on the genes both models fit with a coverage gap on the genes only one model fits, and would substantially overstate the former. The two are reported separately throughout.

One gene cannot be evaluated at all. blaTEM-1 has five positive events in the entire dataset and none in the test split, so no model can be scored on it. The headline macro, nominally over eight genes, is in practice over seven. This is not a property of the gene: the simulator makes blaTEM-1 available only to *E. coli*, although it is in fact common in *Klebsiella pneumoniae* — one survey found it in the great majority of its carbapenemase-producing *K. pneumoniae* isolates [29] — and widening its availability would very likely make it evaluable. We report it as not evaluable rather than reporting a figure.

Per-gene results are not interpretable as biology. As set out in §4.5, per-gene AUROCs are confounded both by sample size, which spans from 19 to 480 positives and runs opposite to the AUROC ordering, and by the seeding mechanism described below. We report them as discrimination measurements only.

Reported spreads are over model seeds, not over data. Every mean and standard deviation in Section 4 varies model initialisation and training order with the dataset held fixed. They therefore describe the stability of training, not sampling uncertainty in the dataset, and they understate total uncertainty — most sharply for the genes with the fewest positive events, where the sampling component is largest and is not represented at all.

5.2 What the simulated biology omits or simplifies

Transfer is intraspecies only. A gene never crosses a species boundary in our simulator, so every cross-species contact is a negative example by construction. This excludes the route that matters most in the literature on these genes: the mcr-1 colistin-resistance plasmid was conjugated into *Escherichia coli* and maintained in *Klebsiella pneumoniae* and *Pseudomonas aeruginosa* [12]; an NDM-1 carbapenemase plasmid was conjugated from *Citrobacter freundii* into an *E. coli* recipient [13]; Tn916-borne tet(M) transferred from *Lactococcus lactis* to *Enterococcus faecalis* both in vitro and in a rat gut [22].

The consequence is one of absence rather than of class balance, and it is worth stating precisely because the intuitive version is wrong. One might expect this restriction to flood the negative class with cross-species pairs that are rejectable from species identity alone. It does not: only 0.275% of contacts in our dataset are cross-species (4,696 of 1,707,498), they occur in only one of the five scenarios, and none of them carries a transfer. Four scenarios contain a single species, and in the mixed scenario the two populations are seeded as separate spatial clusters that seldom come within contact range. What follows instead is that the interspecies case is effectively absent from the data: the model is evaluated almost entirely on within-species pairs, it has not been shown to do anything at all on interspecies transfer, and our metrics cannot register that omission. That is the case of greatest practical interest, and we have no evidence about it either way.

Three of the eleven genes move by a mechanism that is not their real one. We label these simplified mechanisms rather than leave them implicit.

- *gyrA_S83L* is a chromosomal target-site mutation. Quinolone resistance that is genuinely plasmid-borne is carried by a different set of determinants — qnr proteins, a quinolone-modifying acetyltransferase, and mobile efflux pumps — not by mobilised *gyrA* [23, 24]. Our simulator both mutates it into existence, which is right, and transfers it between neighbours, which is not. Additionally, the gene object carries CARD's *Escherichia coli gyrA* entry while being available to three species in the model; we therefore describe it generically as a *gyrA* target-site mutation and treat the specific variant modelled as unverified across those species.
- *acrAB-tolC* is a chromosomal RND efflux system, ubiquitous in the Gram-negative species that carry it, and clinical resistance arises through overexpression rather than through acquiring the genes [25]. The simulator transfers it between cells of a species that already universally possesses it. We note for completeness that plasmid-borne RND pump genes do exist [25], so transferring *an* efflux pump is not in itself unrealistic; what is unrealistic is moving a species' own resident system between its cells.
- *vanA* arises and spreads within MRSA in our model, whereas the documented route is interspecies transfer of Tn1546 from *Enterococcus faecalis* to *Staphylococcus aureus* [26, 27], and *Enterococcus* is not modelled. *vanA* is reported separately from the headline throughout.

These three genes are exactly the ones whose results are most easily over-read: two of them are the rare genes behind the coverage claim in §4.2, and the third posts the highest AUROC of any gene. Their figures measure the learning setup, not resistance biology.

Two further genes are handled by *not* transferring them, which is correct but is itself a simplification: *mexAB-oprM* exists only as an intrinsic gene and so has no events to predict, and *mecA* is marked non-transferable because SCCmec mobilisation is a mechanism the simulator does not implement and conjugation is the wrong substitute for it.

A residual species-level error we did not correct. *Acinetobacter baumannii* carries *mexAB-oprM* and *acrAB-tolC* as intrinsic genes, whereas its characterised RND efflux systems are AdeABC, AdeIJK and AdeFGH [28]; MexAB-OprM belongs to *P. aeruginosa*. We traced the consequences and could identify no effect on the reported numbers — neither gene is in *A. baumannii*'s acquirable set, so neither can spread among those cells, and the only drug in the scenario that features this species is one that *acrAB-tolC* does not cover. We nonetheless record it rather than correcting it silently, because correcting it would change the dataset.

Genes appear from nowhere, and this shapes the dataset. No acquired gene is present at initialisation, so each one enters the population through a mutation branch that adds a uniformly chosen gene from the species' acquirable set with no donor at all. Real acquired genes must come from somewhere; this route exists only to seed the simulation. Because every gene's first carrier arises this way, the number of transfer events per gene is an outcome of the seeding code and of chance rather than of gene biology, as §4.5 sets out — and among three genes matched on species availability and near-matched on transfer probability, event counts still differ roughly two-fold.

Other simplifications carried from the simulator. Cells with no resistance genes receive neither biofilm nor persister protection, because protection is computed from gene-derived resistance before those modifiers are applied; this affects the species that begin with no intrinsic genes. Only bacteria are modelled, so no plasmid host range, no phage, and no environmental reservoir appears. Species composition is fixed per scenario.

5.3 Invented and unvalidated parameters

Several quantities that look like measurements are not, and we label them explicitly rather than let a reader assume provenance from context.

- **Per-gene transfer probabilities** (0.01–0.05 per step) have no cited source, and neither does the rule that a cell attempts transfer on a randomly chosen 30% of its steps.
- **The biofilm conjugation multipliers** are motivated by the general finding that biofilm matrices raise conjugation frequency, but the specific multipliers we use are chosen, not measured.
- **The antibiotic damage function's shape parameters** — a maximum effect of 0.3 and a Hill coefficient of 2 — are hard-coded, with the half-maximal concentration set to each drug's minimum bactericidal concentration. Only the last of these derives from drug data.
- **The dosing protocol** — 0.25 µg/mL cleared after five steps — is an invented protocol. It is expressed in simulation steps because no real duration is assigned to a step.
- **Per-drug decay rates** are unvalidated against pharmacokinetic data. Since no step duration is defined, they cannot even be compared with published half-lives. Assigning a real step duration and sourcing per-drug half-lives is identified future work.

The absence of a defined step duration is the common thread: it means no rate in the simulator can currently be checked against a measured rate. It also bears on the antibiotic-exposure ablation in §4.3, where drug is present for only five of eighty steps, so the finding that exposure features add no measurable signal holds for this protocol and should not be generalised.

5.4 Pipeline history that bears on interpretation

We record two corrections because they change what earlier figures meant, not merely how large they were.

Labels were originally inferred by comparing genomes between consecutive snapshots. That conflates transfer with independent mutation and can attribute a gene's appearance to a neighbour that donated nothing. Labels now come only from events recorded at the moment a transfer fires.

Four model inputs were direct components of the rule that generates the label: the same-species edge flag and the raw edge distance, both hard preconditions for transfer; a transferable-gene count that mirrored the eligibility test; and the SOS flag, the literal term that doubles the transfer probability. A model given these features can partly read the label off its own inputs. All four were removed, taking node features from 36 to 35 dimensions and edge features from 8 to 5. We verified which layout produced the earlier results by rebuilding the architecture under both and matching the parameter counts recorded in the training logs. Figures produced before these two corrections are not comparable with those reported here, and we do not report them. We also note that a file referenced in the source as containing the full leakage audit is not present in the repository, so the account in Section 3.4 is the record.

Finally, a single simulator produced every result in this paper. The task, the feature definitions and the labels all derive from one implementation of one set of modelling decisions, many of them listed above. Performance figures obtained this way describe a model's fit to that implementation. Whether the approach transfers to another simulator, let alone to real populations, is untested, and §5.1's first point applies: no external validation exists yet.

6 Discussion

6.1 What the headline number does and does not demonstrate

A headline AUROC of 0.9805 ± 0.0012 invites a stronger reading than our experiments support, so it is worth being precise about what produced it.

Three of our own results, taken together, suggest that most of the achievable performance comes from the pair of endpoint feature vectors rather than from the surrounding contact structure. Removing message passing entirely costs 0.0031 ± 0.0014 — consistent in direction across every seed, and therefore real, but small enough that a model with no graph convolution at all retains nearly all of the performance. Zeroing feature groups one at a time shows a single group that matters: the genes a cell already carries, at 0.0549 ± 0.0331 , with every other group inside ± 0.003 . And edge features are zeroed in the reported configuration without measurable cost.

The likeliest explanation is that transfer in our simulator requires the donor to carry the gene and the recipient to be able to acquire it, and both conditions are readable from the two endpoint gene vectors. That would account for the genomic group dominating and for the contact graph adding little.

It is worth ruling out a tempting but incorrect second explanation, since we entertained it ourselves. Because transfer is intraspecies only, one might expect the negative class to be padded with cross-species contacts that any model can reject from the species one-hot alone. We measured this and it is not so: only 0.275% of contacts in our dataset are cross-species (4,696 of 1,707,498), they arise in only one of the five scenarios, and none of them carries a transfer. Four of the five scenarios contain a single species, and in the one mixed scenario the two populations are seeded as separate spatial clusters that rarely come within contact range. The intraspecies restriction therefore does *not* make the task easier by supplying easy negatives; what it does instead is make the interspecies case effectively absent from the evaluation, so the model is assessed almost entirely on within-species discrimination (§6.4).

Combined with a held-out split taken over snapshot pairs rather than over runs — so that test windows come from runs the model trained on — the natural reading of 0.98 is that **the task as we posed it is substantially easier than the phrase "predicting horizontal gene transfer" suggests**, not that the model has acquired a deep representation of conjugation. Note that the "easier" here rests on the label structure and the split, not on the class-balance argument we just rejected.

We think this matters beyond our own paper. High discrimination on a simulator-derived task is easy to report and hard to interpret, because the task's difficulty is set by modelling decisions that are invisible in the metric. Our own pipeline supplies the cautionary example: before we removed four inputs that were literal components of the label-generating rule, the same architecture scored higher than it does now. The number improved when the experiment got worse.

6.2 What the results do support: joint training across genes

The result we consider most robust is also the one least dependent on the graph. Our per-gene baselines cannot be fitted for genes with fewer than five positive examples in the training subsample they draw, and because that subsample is drawn per seed, two genes fall below the line in every seed and a third falls below it in three seeds of five. A single model trained jointly on all ten gene outputs produces stable predictions for all of them. That advantage follows from parameter sharing across a multi-label output, not from message passing — the graph-free variant would inherit it too.

This connects to a known weakness of the per-gene approach in the wider literature: a recent review of machine learning for resistance prediction notes that such models typically treat genes as independent predictors [17]. Our setting makes the cost of that assumption concrete and measurable, because the rare targets are exactly the ones a per-gene model cannot reach. The mechanism is mundane — shared representations let common genes subsidise rare ones — but for a problem where the interesting determinants are often the rare ones, it is the part of our design we would keep.

Two caveats travel with it, and neither is optional. The genes concerned have 19, 34 and 94 positive events in the whole dataset, so their AUROCs are imprecise in a way that seed-to-seed standard deviations do not capture. And two of the three are chromosomal in real bacteria and are moved by our simulator as a deliberate simplification, so for those the result demonstrates a property of the learning setup and not a fact about the genes. The third, blaKPC-2, carries no mechanism caveat, which makes it the cleanest case: a genuine plasmid-borne gene that the per-gene baseline fits in two seeds out of five, returning 0.9611 and 0.8179 where it does fit, against a jointly trained model that returns 0.9564 $\pm$ 0.0021 every time.

6.3 Relation to prior approaches

The closest prior work predicts horizontal transfer between *genomes*: Zhou et al. infer recent transfer events from near-identical sequence shared by distantly related organisms and predict the resulting network from functional gene content, finding that a random forest on functional profiles performs best and that a graph convolutional network performs comparably, improving as network topology is added [19]. Recent graph neural network work on resistance operates at the isolate or patient level, classifying nodes that are whole isolates connected by genomic distance [20].

Our formulation sits at a different scale — individual cells, their momentary contacts, and a specific next time window — and the trade is explicit. Genome-level work has real data and indirect labels, inferred from sequence similarity over evolutionary time. We have exact labels and simulated data. Neither is a substitute for the other, and our numbers are not comparable with theirs: different unit of prediction, different notion of an event, different data. We draw no comparison of magnitudes anywhere in this paper.

It is worth noting that the direction of our own ablation echoes theirs. In their setting a non-graph model matched or beat the graph model on functional features; in ours, removing message passing costs 0.003. Two studies at different scales both finding limited benefit from graph structure is weak evidence, but it points the same way, and we would rather record it than present message passing as more load-bearing than it is.

6.4 The interspecies gap

The restriction we consider most limiting is that a gene never crosses a species boundary in our simulator. The literature on the very genes we model is largely about transfers that do: the mcr-1 plasmid conjugated into *Escherichia coli* and maintained in *Klebsiella pneumoniae* and *Pseudomonas aeruginosa* [12], an NDM-1 plasmid conjugated from *Citrobacter freundii* into *E. coli* [13].

This is not a gap that a better model closes. It is a gap in the simulator, and it propagates into the task, the labels and the evaluation: a model trained where cross-species contacts are always negative has had no opportunity to learn anything about the case of greatest practical interest, and our metrics cannot detect that it has not. Adding interspecies transfer means deciding, per gene, which species boundaries

it can cross and at what rate — which is a modelling problem requiring sourced host-range information rather than a parameter to tune. We regard it as the most valuable extension, and the one most likely to reduce the headline numbers, which we would treat as the experiment working rather than failing. Section 7.3 sets out what it would take.

6.5 On calibration and precision–recall

Two secondary results deserve a brief, honest reading. Predicted probabilities are well calibrated (expected calibration error 0.0012 ± 0.0003), which is useful if such a model were ever to inform a ranking, but calibration on a simulator's own events says nothing about calibration against reality. Headline AUPRC is far above the base rate yet has a standard deviation nearly equal to its mean, so we treat it as evidence that the ranking carries information and not as a stable performance figure. Under extreme class imbalance the precision–recall summary is dominated by the handful of genes with the fewest positives, which is precisely where our dataset is least informative.

Finally, the finding that local antibiotic exposure adds no measurable signal should not be read as a claim about selection pressure. Under our protocol the drug is present for five of eighty steps. The honest conclusion is that these features carry no signal in this protocol, and that a protocol with sustained exposure would be needed to say anything more.

7 Future Work

Each item below states what it would change about this paper's claims and roughly what it costs. Items 1 and 5 are the two we would do first, for different reasons: item 1 is cheap and changes how every number should be read, and item 5 removes a simplification that currently undercuts the result we most want to stand.

Two costs recur, so we state them once. A full regeneration and retrain of the reference result set took **3,510 s (about one hour)** on a single CUDA GPU, as recorded in the results file, plus a few minutes of data generation. That is cheap enough that none of these items is blocked by compute; what they cost is design work, literature work, and in three cases the invalidation of every number in Section 4. Person-day figures are our estimates, not measurements.

7.1 Split by run or by scenario, not by snapshot pair

What it changes. This is the only item that changes the paper's numbers without changing the simulator, and the only one that changes what those numbers *mean* rather than what they measure. Every figure in Section 4 is currently computed on a split over snapshot pairs, so it reports generalisation to unseen time windows of runs seen in training (U13). A grouped split would replace S1–S9 with figures that answer the question a reader assumes is being answered. We expect them to be lower; that is the point.

What it takes. The code change is small. `split_dataset` shuffles pairs; it would group by run instead. The run identity is not currently in a graph's metadata — which carries step, scenario, node and edge counts — so `collect_training_snapshots` would need to stamp each pair with its scenario and data seed first. That is perhaps half a day.

The real obstacle is that fifteen runs is too few to group. Five scenarios $\times$ three data seeds, split 70/15/15 by run, leaves about ten training runs and three test runs, which is too coarse for a stable

estimate and would make the reported standard deviations largely a statement about which three runs landed in the test fold. Doing this properly means generating more runs — on the order of ten data seeds per scenario, so fifty runs — and reporting either leave-one-scenario-out (five folds, which also measures transfer to an unseen population composition) or leave-one-seed-out. At roughly ten seconds per run, the extra data generation is minutes; the retraining is five folds $\times$ five model seeds, so a handful of hours at the recorded rate.

Estimated effort: 2–3 person-days plus a few hours of compute. **Consequence:** S1–S9 are superseded; U13 is resolved.

7.2 Per-gene transfer mobility

What it changes. The simulator applies one mechanism — conjugation between same-species neighbours — to every gene, regardless of how that gene actually moves. Fixing this would resolve U10 and the vanA half of U6, and would let per-gene results be read as something other than an artefact.

It would also remove the demonstration behind our coverage claim, which is why we rank it second. Two of the three genes in S2b, *acrAB-tolC* and *gyrA_S83L*, are exactly the ones whose transfer is a simplification. (The third, *blaKPC-2*, is unaffected and would survive the change, which softens this consequence but does not remove it.) If *gyrA_S83L* becomes mutation-only and *acrAB-tolC* becomes regulation-only — which is what the sources in §5.2 imply — then both drop out of the transfer label set, and the finding that joint training reaches genes the per-gene baselines cannot fit needs a new demonstration. *blaTEM-1* is the natural candidate (5 positives, currently not evaluable; see item 6), but the claim would have to be re-earned rather than carried over. We would rather discover that now than have a reader discover it later.

What it takes. The design work is deciding what mechanisms to model and how each one moves: a conjugative plasmid (what the model already does), a conjugative transposon such as the Tn916 that carries *tet(M)*, a chromosomal point mutation that should not transfer at all, a regulatory-overexpression route for resident efflux systems, and a mobile genomic island such as SCCmec for *mecA*. Each needs a rate and a host range with a source rather than a chosen number. The code change is contained — `_attempt_hgt` already consults a per-gene `non_transferable` set, so the extension is a per-gene mechanism tag and a branch per mechanism.

Estimated effort: 1–2 person-weeks, most of it sourcing per-gene mechanism parameters, plus a regeneration. **Consequence:** U10 resolved, U6 partly resolved, S2b needs re-demonstration, all of Section 4 regenerated.

7.3 Interspecies transfer

What it changes. This is the largest scope restriction in the paper. Transfer is currently intraspecies only, so the interspecies case that dominates the literature on these genes is absent from the task, the labels and the evaluation (U11), and our measurements cannot detect that absence — only 0.275% of contacts in our dataset even cross a species boundary (S10). Adding it would make the task match the phenomenon the paper names, and would very likely lower the headline figures, since the model would face a class of positives it has never seen.

What it takes. Not a parameter sweep. It requires deciding, per gene, which species boundaries it can cross and at what relative rate, which is a literature question: plasmid incompatibility groups and documented host ranges, the kind of evidence cited in §5.2 for *mcr-1*, NDM-1 and *tet(M)*. It also

interacts with scenario design, because four of the five current scenarios contain a single species and the fifth seeds its two species as separate spatial clusters that seldom touch (S10) — so scenarios with genuinely mixed, spatially interleaved populations would be needed for interspecies contacts to be common enough to learn from. That is a scenario-design change as much as a mechanism change.

Estimated effort: 2–3 person-weeks, dominated by host-range sourcing and scenario redesign, plus a regeneration. **Consequence:** U11 resolved; S10 no longer describes the dataset; every figure in Section 4 regenerated on a harder task.

7.4 External validation against real gene calls

What it changes. U7 currently says no external validation exists. Our attempt failed for a data reason: the isolate collections we obtained carry resistance phenotypes, not per-isolate gene calls, so gene prevalences collapse and the rank correlation is undefined. Re-annotating assemblies with a resistance-gene caller such as ResFinder or AMRFinderPlus would supply the gene-level ground truth the pipeline needs.

What it would and would not validate, which we think matters more than the effort. A static collection of annotated genomes contains no transfer events, so it cannot validate the predictor. What it can validate is the *simulator*: whether the gene prevalences and co-occurrence patterns our populations produce resemble those in real isolates of the same species. That is worth doing — it is the difference between a simulator nobody has checked and one whose output distributions have been compared against reality — but it should be reported as validation of the data-generating process, not of the model, and we would be explicit about that. Validating the *predictions* would need observed transfer events, which means either longitudinal sampling with strain tracking or transfer networks inferred from sequence similarity, as in the genome-level work discussed in §6.3. We regard that as a separate study.

Estimated effort: about 1.5 person-weeks for the simulator-distribution comparison, matching our earlier scope for the re-annotation work, and substantially more for anything that validates the predictor. **Consequence:** U7 partly resolved, with the scope of the resolution stated precisely.

7.5 A defined step duration, and sourced pharmacokinetics

What it changes. No simulated step is assigned a real duration anywhere in the code, and that single omission blocks every pharmacological comparison: per-drug decay rates cannot be checked against measured half-lives, the damage function's shape parameters cannot be checked at all, and the dosing protocol is necessarily arbitrary rather than a modelled course of treatment (§5.3). Fixing it would let several invented parameters be replaced with sourced ones.

What it takes. A calibration decision and then literature work. The natural anchor is growth: the model already assigns each species a doubling time in steps, so fixing a real doubling time for one species fixes the step. That choice is itself a claim needing a source and a stated growth condition, since doubling times vary widely with medium and temperature. Given a step duration, per-drug half-lives convert to per-step decay rates directly, and the dosing protocol can be restated as a real concentration over a real interval. We would treat the whole item as a biology change requiring sourced values rather than chosen ones.

Estimated effort: 3–5 person-days of sourcing and calibration, plus a regeneration. **Consequence:** several entries in §5.3 move from invented to sourced; the antibiotic-exposure result (U4) becomes

testable under a realistic course rather than one lasting five of eighty steps.

7.6 A seeding mechanism with a biological counterpart

What it changes. Acquired genes currently enter the population through a mutation branch that adds a random gene from a species' acquirable set with no donor at all. Nothing in reality corresponds to this, and it is the reason the per-gene positive counts cannot be interpreted (U12) and the reason blaTEM-1 is not evaluable (U8). Replacing it — with a small seeded carrier population, or an environmental reservoir from which genes are acquired — would make the per-gene counts a modelled quantity rather than an artefact, and would let the counts be set deliberately instead of falling out of pool sizes and chance.

What it takes. Small code change, real design thought about what the reservoir represents, and a regeneration. It pairs naturally with item 2, since both concern how genes enter and move.

Estimated effort: 3–5 person-days plus a regeneration. **Consequence:** U12 resolved; U8 likely resolved; S9 becomes interpretable; per-gene counts become a designed property of the experiment.

7.7 Summary

#	Item	Effort (estimated)	Claims affected
1	Group the split by run or scenario	2–3 days + hours of compute	S1–S9 superseded; U13 resolved
2	Per-gene transfer mobility	1–2 weeks + regeneration	U10 resolved; U6 partly; S2b needs re-demonstration
3	Interspecies transfer	2–3 weeks + regeneration	U11 resolved; S10 obsolete; Section 4 regenerated
4	External validation via gene calls	~1.5 weeks (simulator only)	U7 partly resolved, scope stated
5	Defined step duration + sourced PK	3–5 days + regeneration	§5.3 entries sourced; U4 testable
6	Seeding with a biological counterpart	3–5 days + regeneration	U12 resolved; U8 likely; S9 interpretable

Items 2, 3, 5 and 6 all change the simulator and therefore invalidate every figure in Section 4; they would sensibly be batched into one regeneration rather than run separately. Item 1 is independent of all of them and should be done first, because until it is, none of the reported figures answers the question a reader is most likely to think it answers.

8 Conclusion

We posed horizontal gene transfer prediction as a supervised problem at the level of individual cells: given a snapshot of a simulated bacterial population, predict for each directed cell-to-cell contact and each resistance gene whether that gene is transferred in the next time window. Because the labels are events the simulator records as they fire, the ground truth is exact — which is the property that motivated the formulation and is not available from observational genomic data.

On this task a graph attention network trained jointly on all gene outputs reaches a test AUROC of 0.9777 ± 0.0023 on the four genes our per-gene random forest baseline fits in every seed, against

0.9548 $\pm$ 0.0124 for that baseline, and is ahead on every seed. It also gives stable predictions for three further genes the baselines fit unreliably or not at all — an advantage that comes from sharing parameters across a multi-label output rather than from the graph.

That last distinction is the paper's main methodological conclusion. Message passing contributes a small but consistent 0.0031 $\pm$ 0.0014, and the genes a cell already carries dominate every other feature group by an order of magnitude. On this task, most of the signal is available from the two endpoint feature vectors, and what earns the architecture its place is joint training across genes, not the propagation of information over the contact graph. We report that rather than the more flattering alternative reading.

The scope of these numbers is narrow, and deliberately stated as such throughout. They are internal to one simulator; no external validation exists, and our attempt at one failed for want of gene-level calls in the data we could obtain. The simulator transfers genes only between cells of the same species, so the interspecies transfers that dominate the literature on these very genes are absent from the task, the labels and the evaluation alike. The held-out split is taken over snapshot pairs rather than over runs, so it measures generalisation to unseen time windows of seen simulations and not to unseen simulations. Three of the eleven genes are moved by a mechanism that is not their real one, and several parameters that resemble measurements are invented. Under those conditions, a high AUROC is better read as a statement about the difficulty of the task as posed than as evidence of a model that understands conjugation.

What we think survives those caveats is the formulation and one design choice: an exactly labelled, per-contact, per-gene transfer task is a usable testbed, and training one model across all genes reaches rare targets that per-gene models cannot. The pipeline that produces these results is seeded and deterministic across processes, with a frozen configuration guarded byte-for-byte by a regression test, so the numbers can be reproduced and, more importantly, can be moved by the changes set out in Section 7: splitting by run, adding interspecies transfer, modelling per-gene mobility, and validating against real gene calls. We expect several of those changes to lower the figures reported here, and regard that as the point of making them.

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Appendix A. Citation notes

Every reference above was checked against the source itself, not merely confirmed to exist: author list, venue, year and DOI, and whether the source actually supports the claim it is cited for. The audit is recorded in `paper/citations.md` and, for gene transfer mechanisms, in `paper/gene_mechanism_audit.md`.

Because this version has no submission deadline, sources whose full text sits behind a paywall were verified against their abstracts and are cited only for what those abstracts state. They are listed here so the distinction is on the record rather than implied.

#	Reference key	Verified against
1	gbd2019amr	publisher page and abstract
2	gbd2021amr	publisher page and abstract
3	vonwintersdorff2016	abstract
4	partridge2018	abstract
5	seoane2011	abstract
6	merkey2011	abstract
7	brito2021	abstract
8	yaffe2020	full text
9	hellweger2016	abstract
10	krone2007	abstract
11	velickovic2018	preprint page and abstract
12	liu2016mcr	abstract
13	dolejska2012	abstract
14	lardon2011	abstract
15	gorochowski2012	abstract
16	glushchenko2019	abstract
17	kim2022	abstract
18	ellabaan2021	abstract; its 2026 Author Correction read in full
19	zhou2021	full text
20	nguyen2026amrgnn	full text
21	donabauer2025	abstract
22	boguslawska2009	abstract
23	hooper2015	abstract
24	strahilevitz2009	abstract
25	li2015efflux	abstract
26	weigel2003	abstract
27	clark2005	abstract
28	coyne2011	abstract
29	cuzon2010	abstract

23 of the 29 cited sources were verified at abstract level only. Upgrading them to full text is listed as outstanding work in `paper/STATUS.md`.

`paper/refs.bib` additionally holds `bonnin2012`, `dandrea2013`, `devirgiliis2009` — sources verified during the gene-mechanism audit that the text does not cite. They are kept for that record and are deliberately absent from the reference list above.